

INVENTIVE MINDS

A JOYFUL YEAR-LONG JOURNEY INTO THE WORLD OF TECHNOLOGY AND CREATIVITY



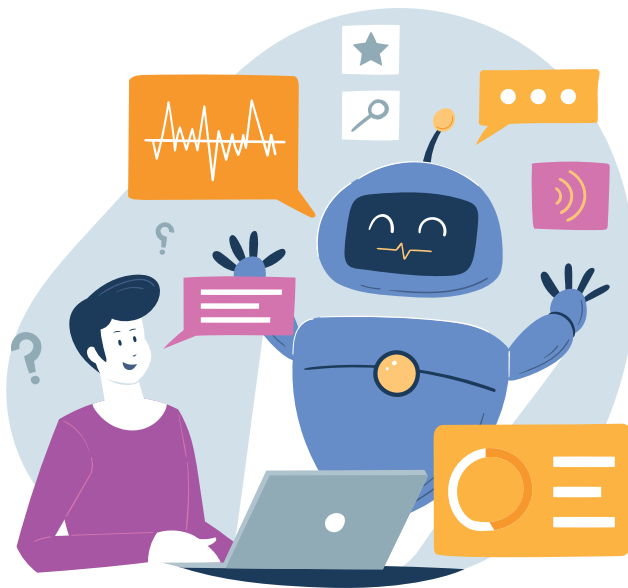
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GRADE



Build, and improve AI- and robotics-based smart systems using rules and data in mBlock. They combine multiple parts into working solutions while developing problem-solving and presentation skills.

Grade – 1

INVENTIVE MINDS



TECH EXPLORERS

Inspiring Future Innovators

Grade – 1

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Preface

At VeriTon, learning begins with understanding how things work.

Young learners are naturally curious. They observe, ask questions, and explore the world around them. This book guides that curiosity by helping students recognise simple systems in everyday life.

Students learn that every system follows a pattern:

Input → Process → Output

Through real-life examples and guided activities, they understand how actions start, how changes happen, and how results are produced.

Learning follows a simple path:

Observe → Understand → Try → Improve

Students first observe, then build understanding, and finally apply their ideas through simple activities. Technology is used only as a support tool. The focus remains on thinking, exploring, and making sense of systems. This book introduces learners to four connected areas:

Digital Modelling — exploring ideas safely before building

Robotics Smart Systems — recognising how devices respond to signals

Coding — understanding instructions and sequences

Artificial Intelligence — using digital tools as assistants

The aim is to help students become thoughtful learners who understand systems, not just use technology.

Key Features

1. Digital Design and Virtual Modelling

Students learn how ideas can be tested using digital models before building real systems.

They explore simple virtual circuits and understand how real objects are represented on a computer.

2. Signals and Smart Objects Around Us

Students observe how everyday objects respond to signals such as buttons, light, and touch.

They understand how devices perform actions based on input and how sensors enable automatic behaviour.

3. Coding Through Instructions and Actions

Students learn how computers follow instructions.

They explore sequence, triggers, and repetition to create simple movements and behaviours in digital characters.

4. Digital Creativity and Story Building

Students create simple animations using characters, backgrounds, movement, and sound.

They learn how visual elements and actions combine to express ideas.

5. AI Tools for Learning and Language

Students use AI tools to improve sentences and translate words.

They understand that AI supports learning but must be used carefully and checked by humans.

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My First Digital Model

Objectives

1. Understand how digital designs help us test ideas.
2. Identify basic electronic parts such as LED, battery, and switch.
3. Understand how light and switches work in a virtual circuit.
4. Learn how electricity flows in a simple circuit.
5. Build a simple digital model using an online design tool.



1. How design works:

Real-life example

Before building a real house, architects first create drawings and models.

These designs help them test ideas safely.

In the same way, engineers often design circuits on a computer first.

This helps them understand how things will work.



1.1 What is design

Definition

Design means planning how something should look or work.

People often draw or model their ideas before building them.

This helps them test ideas and fix mistakes.

Design makes building easier and more organised.



Example

A student designs a lamp model before building a real one.

1.2 Real objects and digital models

Definition

A digital model is a computer version of a real object.

It shows how the object looks or works.

Students can experiment with the model safely.

Digital models help us learn about real systems.



Example

A digital LED model represents a real LED light.

Summary

- » Design helps us plan ideas before building.
- » Digital models represent real objects on a computer.

Let's think about this

Teacher shows a real torch.

Students identify the parts inside the torch such as the battery, switch, and light.

Did You Know?

Engineers often design machines and circuits digitally before building them in real life.



About the tool: Tinkercad Circuits

What this tool does

Tinkercad Circuits is an online design tool.

It allows students to build electronic circuits on a computer.

The tool provides virtual parts such as batteries, LEDs, and switches.

Students can connect these parts and test how the circuit works.

How it works

Students drag components into the workspace. They connect the components using virtual wires. The program simulates electricity flow in the circuit. When the circuit is complete, the device starts working.



Activity 1

Virtual vs Real LED Comparison

Challenge Scenario

In class, we can see a small LED light. On the computer, we see a digital LED. Are they the same or different?

Goal of the Activity

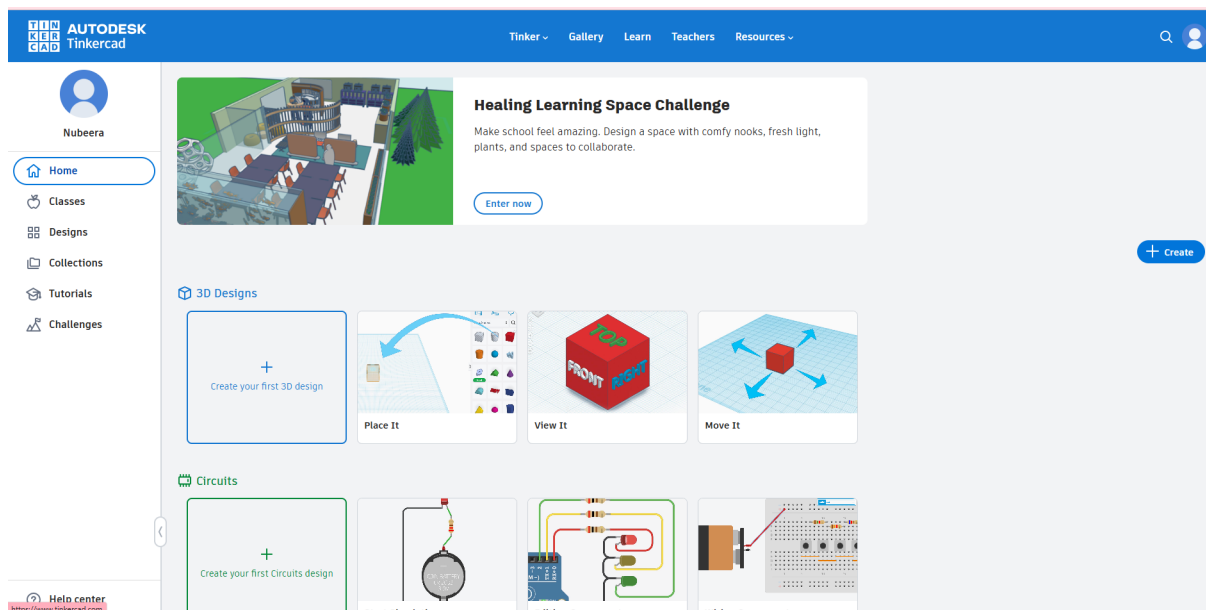
We will look at a digital LED in Tinkercad and learn how it works.

Materials Required

» LED

Simulation Setup

Step 1: Access the Platform



Step 2: Arrange Components

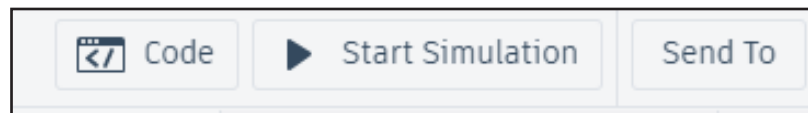
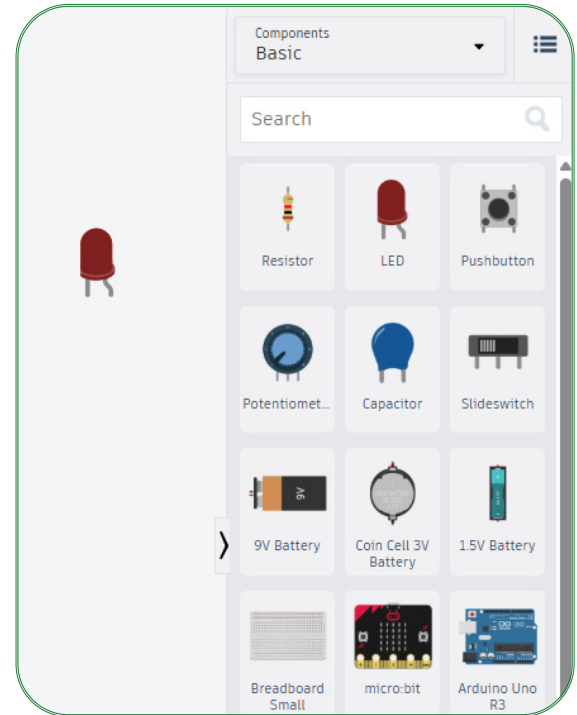
- » Find “LED” in the components panel
- » Drag the LED into the workspace
- » Place it in the centre

Step 3: Build the Circuit

- » We will not connect anything in this activity
- » Just look at the LED

Step 4: Run the Simulation

- » Click Start Simulation
- » Watch what happens
- » Click Stop Simulation



Conclusion

The LED on the screen is like a real LED, but it will only glow when we connect it properly.

Key Learnings

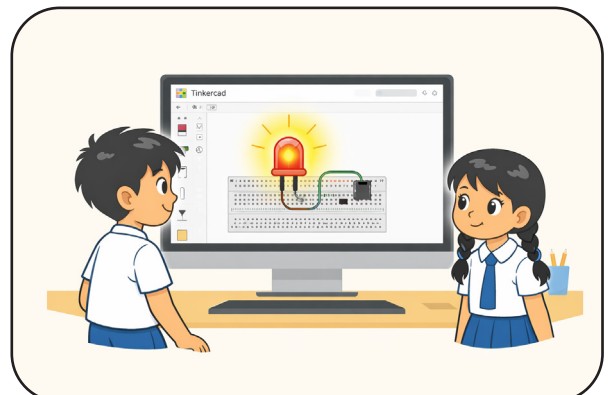
- » A digital LED is like a real LED
- It needs connections to glow
- » We can learn safely using Tinkercad

2. Light and switch in virtual space

Real-life example

When we press a lamp switch, the light turns on. The switch controls the electricity flowing to the bulb.

In a digital circuit, the same idea works using virtual



parts.

2.1 What is an LED

Definition

An LED is a small electronic light.

It glows when electricity flows through it.

LEDs are used in toys, indicators, and electronic devices.

They are commonly used in simple circuits.

Example

An LED lights up when connected to a battery.

2.2 What is a switch

Definition

A switch controls the flow of electricity in a circuit.

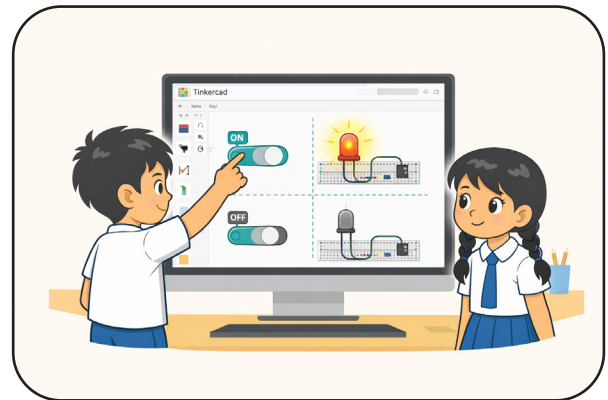
It can turn the circuit on or off.

When the switch is closed, electricity flows.

When the switch is open, electricity stops.

Example

Pressing the switch turns the light on.

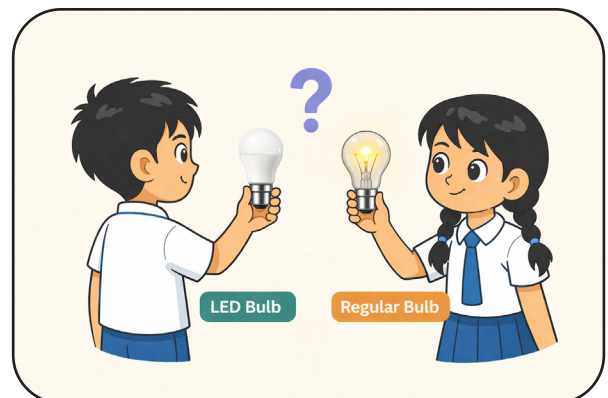


Summary

- » An LED produces light in a circuit.
- » A switch controls the light.

Did You Know?

LED lights use less electricity and last longer than many other types of bulbs.



Activity 2

Build a Virtual LED Circuit

Challenge Scenario

We want to make a small light glow on the computer. How can we do this using Tinkercad?

Goal of the Activity

We will build a simple LED circuit in Tinkercad and make the LED glow.

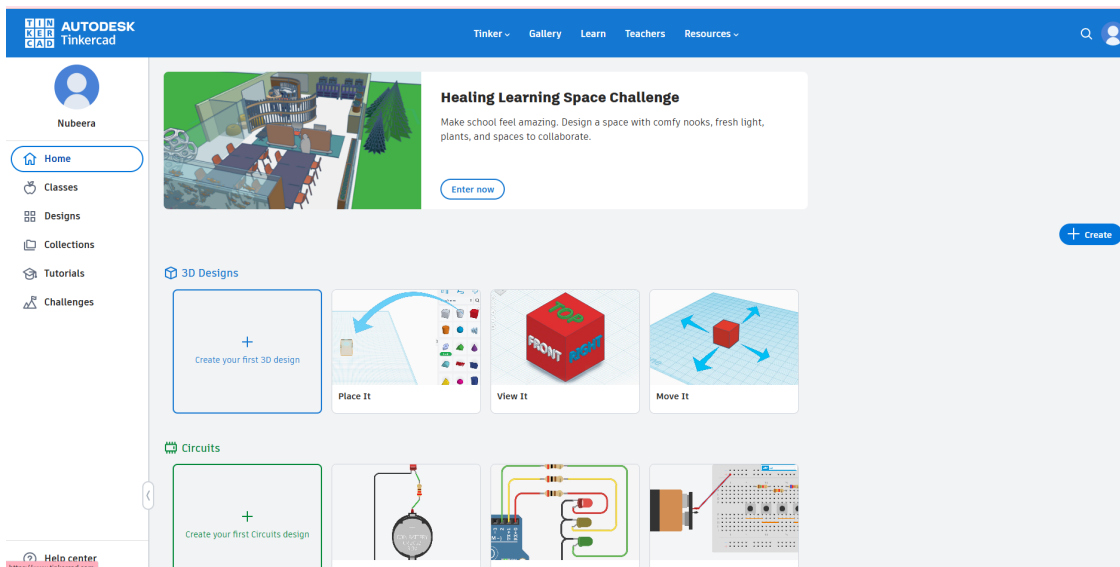
Materials Required

- » LED
- » Battery
- » Register

Simulation Setup

Step 1: Access the Platform

- » Open a web browser
- » Go to <https://www.tinkercad.com>
- » Click on Circuits
- » Click on Create New Circuit



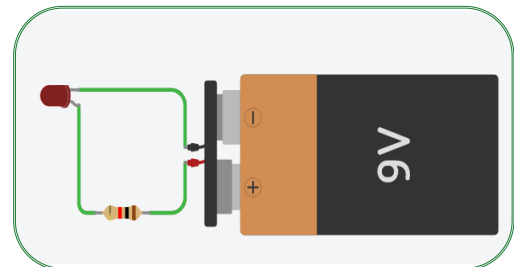
Step 2: Arrange Components

- » Find “LED” in the components panel
- » Drag it into the workspace
- » Find “Battery” and drag it into the workspace
- » Find “Resistor” and drag it into the workspace
- » Place both neatly



Step 3: Build the Circuit

- » Click on the battery positive (+) terminal
- » Drag a wire to one end of the **resistor**
- » Connect the other end of the resistor to the **long leg (anode) of the LED**
- » Click on the short leg (cathode) of the LED
- » Drag a wire back to the battery negative (–) terminal

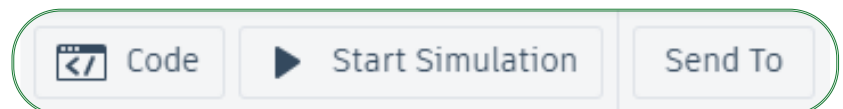


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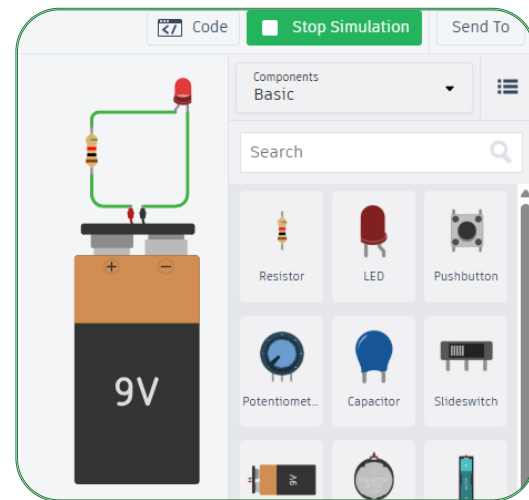
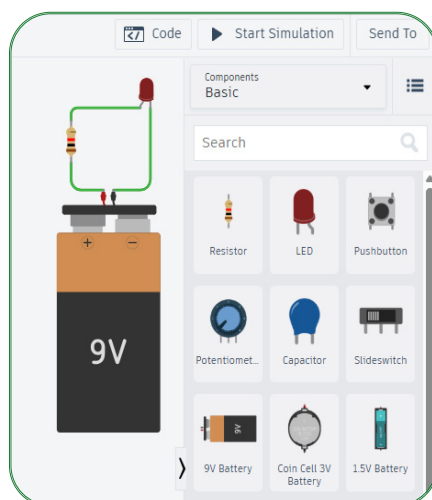
The resistor is important because it limits the current and prevents the LED from burning out.

Step 4: Run the Simulation

- » Click Start Simulation
- » Watch the LED glow
- » Click Stop Simulation



Before and After Simulation :



Conclusion

The LED glows because we made a complete path for electricity using the battery and wires.

Key Learnings

- » An LED needs a power source to glow
- » Wires help make a complete path
- » A circuit must be complete to work

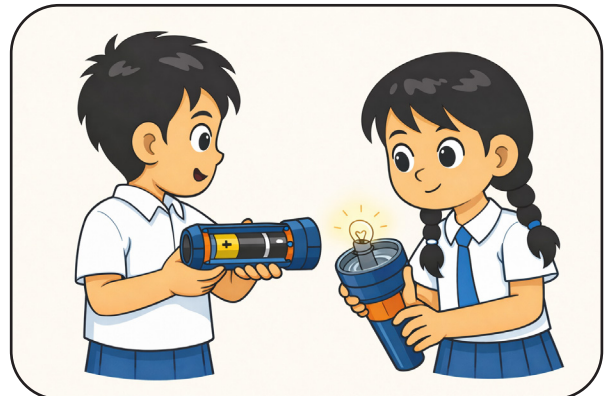
3. Understanding basic circuit flow

Real-life example

A torch lights up because electricity flows from the battery to the bulb.

If the connection breaks, the light stops.

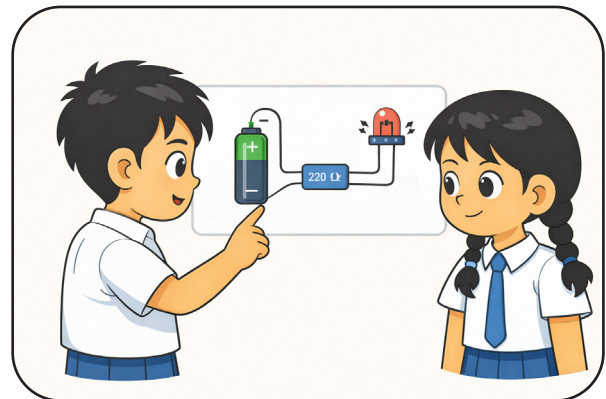
Electricity always needs a path to travel.



3.1 What is a circuit

Definition

A circuit is a path through which electricity flows. The path connects the battery and electronic parts. Electricity moves through this path to power devices. If the path breaks, the circuit stops working.



Example

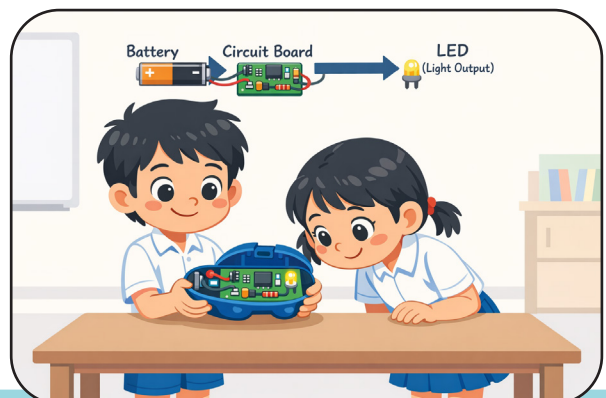
A battery connected to an LED creates a working circuit.

Summary

- » Electricity flows through a circuit.
- » A complete circuit allows devices to work.

Did You Know?

Every electronic device in your home works because of circuits inside it.



Activity 3

Add a Switch to Control Light

Challenge Scenario

Our LED is always on. How can we turn it on and off when we want?

Goal of the Activity

We will add a switch in Tinkercad to control the LED.

Materials Required

- » LED
- » Battery
- » Switch
- » Resistor

Simulation Setup

Step 1: Access the Platform

- » Open a web browser
- » Go to <https://www.tinkercad.com>
- » Click on Circuits
- » Open your saved circuit or click Create New Circuit

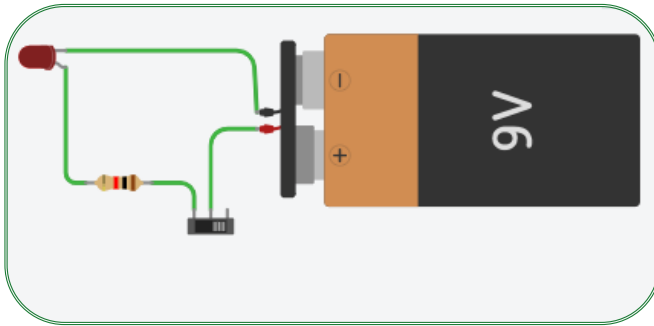
Step 2: Arrange and Connect Components

- » Click on the battery positive (+) terminal
- » Drag a wire to one terminal of the switch
- » Connect the other terminal of the switch to one end of the resistor
- » Connect the other end of the resistor to the long leg (anode) of the LED
- » Click on the short leg (cathode) of the LED
- » Drag a wire back to the battery negative (–) terminal

This keeps everything

in a proper series circuit: **Battery → Switch → Resistor → LED → Battery**

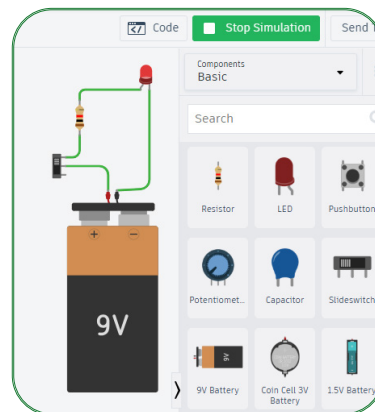
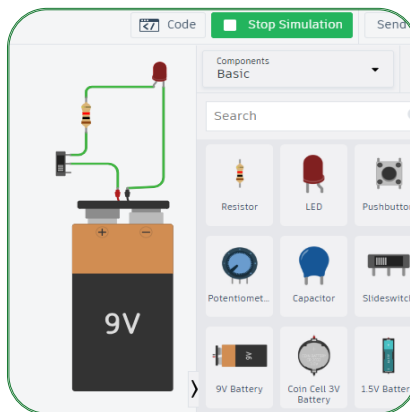
After Connecting wires :



Step 3: Run the Simulation

- » Click Start Simulation
- » Click the switch to turn it ON and OFF
- » Click Stop Simulation

Switch off and on mode :



Conclusion

The switch controls the flow of electricity. When the switch is ON, the LED glows. When it is OFF, the LED turns off.

Key Learnings

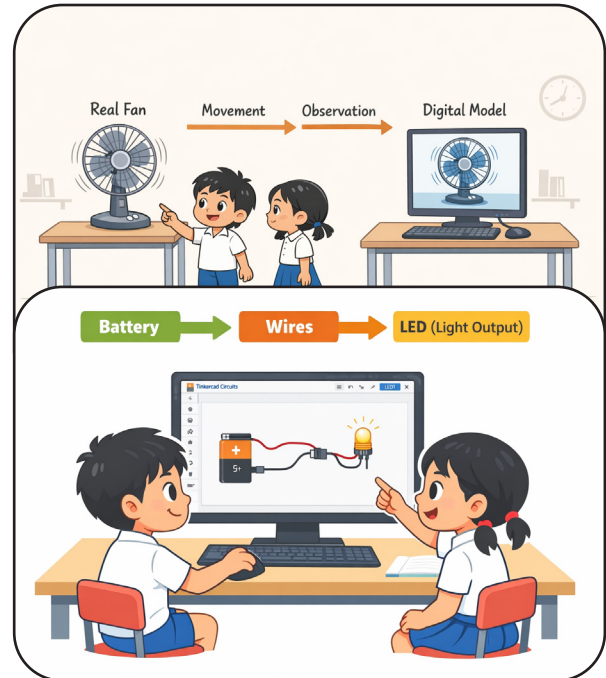
- » A switch can control a circuit
- » Electricity flows only when the switch is ON
- » We can turn devices on and off safely

4. Creating simple digital models

Real-life example

Electric fans use electricity to spin their blades. A motor inside the fan makes the blades move.

We can design a simple fan model in a digital circuit tool.



4.1 Designing a simple digital model

Definition

“Digital models are created using virtual parts.”
Students connect components to design systems.
The tool shows how the system behaves.
Digital models help test ideas safely.

Example

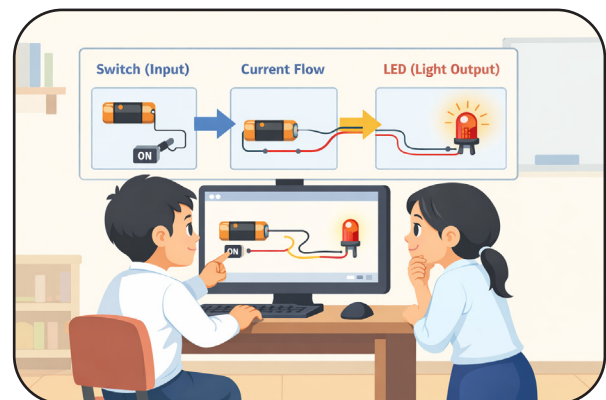
A digital fan model shows how electricity powers movement.

Summary

- » Digital models help us test design ideas.
- » Virtual circuits allow safe experimentation.

Did You Know?

Many engineers test their designs on computers before building real machines.



Activity 4

Build a Virtual Fan

Challenge Scenario

We use a fan to feel cool. Can we make a fan work on the computer using Tinkercad?

Goal of the Activity

We will build a simple circuit to make a motor spin like a fan.

Materials Required

- » Battery
- » Switch
- » DC Motor

Simulation Setup

Step 1: Access the Platform

- » Open a web browser
- » Go to <https://www.tinkercad.com>
- » Click on Circuits
- » Click on Create New Circuit

Step 2: Arrange Components

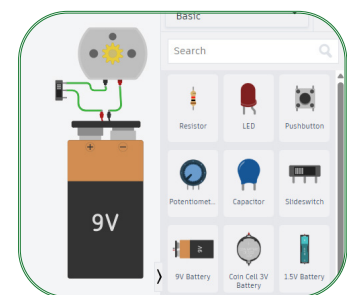
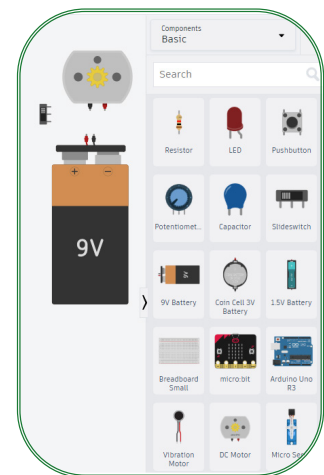
- » Find “Battery” and drag it into the workspace
- » Find “Switch” and place it near the battery
- » Find “DC Motor” and place it next to the switch
- » Arrange all parts neatly

Step 3: Build the Circuit

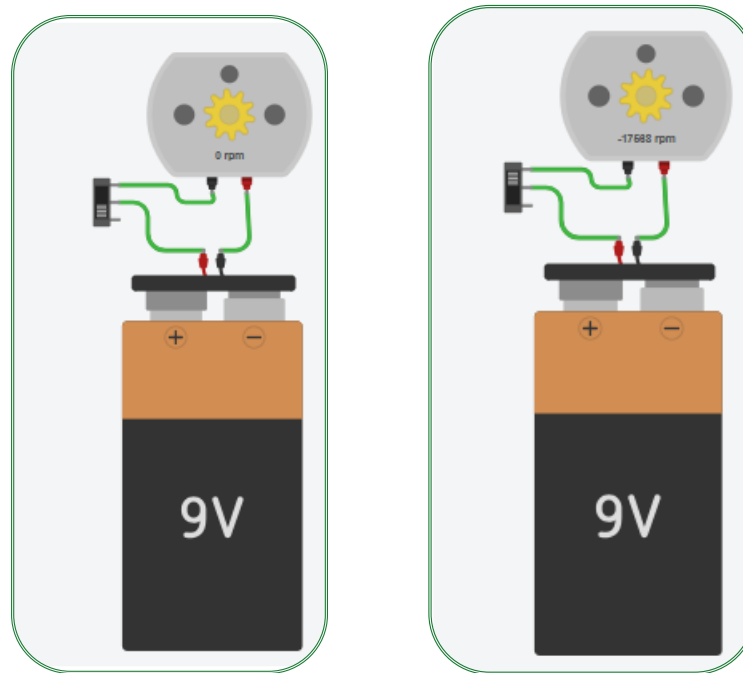
- » Connect battery positive (+) to one side of the switch
- » Connect the other side of the switch to one terminal of the motor
- » Connect the second terminal of the motor to battery negative (–)

Step 4: Run the Simulation

- » Click Start Simulation
- » Click the switch to turn it ON
- » Watch the motor spin
- » Click Stop Simulation



Switch off and on Mode:



Conclusion

The motor spins when electricity flows through the circuit. The switch helps us control when the fan turns on and off.

Key Learnings

- » A motor can spin when it gets power
- » A switch can control the motor
- » We can build a simple fan using a circuit